

CineMatch Sprint 2 Report: Exploring a Big Movie Dataset with UNIX
Team 6 (CineMatch) · CS 131, Section 02 · September 28, 2026

1. Stakeholder Q&A

Who cares: The engineer building the CineMatch's movie recommendations.

The Question: The IMDB platform currently lists the people who have worked on a title before. Every listing on the dataset lists out two ways to describe what someone did. This includes a category (such as actor) and this also includes a job (which is a typed out job title). Which category should the system use?

Proposed Answer: We propose to use categories. This is because over 80% of the 100+ million listings don't have a job filled in; on the other hand every single listing has a category filled in.

2. The dataset

We used IMDB free dataset (<https://datasets.imdbws.com/>). There are 7 files in our team folder: /mnt/scratch/CS131_jelenag/projects/team06_sec02_fall2026/imdb_data/.

Size check:

- All 7 files together are over 10 GB with over 200 million lines.
- This meets both of the requirements with having over 10GB and having over 50 million rows. We measured this with:

The file we analyzed: title.principals.tsv.

- It has over 100 million lines and it includes a header line with column names and over 100 million listings.
- The columns are also separated by tabs which is convenient

3. What we found

Table 1: Includes the listings by the category

```
tail -n +2 title.principals.tsv | cut -f4 | sort | uniq -c | sort -nr
```

Listings	Role type
24,083,232	actor

18,198,668	actress
15,495,183	self
12,114,558	writer
8,609,438	director
7,520,205	producer
5,526,847	editor
4,088,847	cinematographer
3,231,567	composer
1,189,576	production_designer
1,166,912	casting_director
653,639	archive_footage
14,136	archive_sound

What it means: This shows how over half the listings include people such as actors/actresses and people playing themselves. Since the 13 numbers add up to 101,892,808 this proves that every listing stated has a role type.

Table 2: listings by job title (job), top 10 result

```
tail -n +2 title.principals.tsv | cut -f5 | sort | uniq -c | sort -nr
```

Listings	Job title
82,252,330	\N (blank)
7,336,567	producer
1,861,627	writer
1,073,947	director
820,259	written by
765,108	editor
684,417	creator
656,381	created by

654,170	composer
630,525	director of photography

This means that the most common job title in this table shows as blank, and when it's filled in this shows that the same job gets in in different ways. For example the words can be “writer” or “written by” or “creator” or “created by”. This means we likely need to normalize the job title because currently it's unreliable.

Top 10 people with the most listings

```
tail -n +2 title.principals.tsv | cut -f3 | sort | uniq -c | sort -nr | head -n 10
```

Listings	Name	What they do
39,896	Ekta Kapoor	writer, producer, director
32,510	Snehasish Chakraborty	writer, producer, composer
32,475	Shobha Kapoor	producer, writer
30,142	Saibal Banerjee	producer, director, writer
28,452	Leena Gangopadhyay	writer, producer
26,781	Reg Watson	writer, producer, director
25,370	Michael Egizi	composer
23,751	Johnny Manahan	director, writer
22,224	Mark Goldbridge	producer, director
21,993	Aashish Rego	composer, producer

This shows how none of the top 10 people are actors. Most of them are actors. They're mostly producers and writers of long-running daily TV shows, where one person gets listed on every episode. So having the most listings doesn't mean someone matters most for movies. We'll need to filter movies only before ranking people.

Filter: how many listings have a blank job?

```
tail -n +2 title.principals.tsv | cut -f5 | grep -E -x '\N' | wc -l
```

The result showed that 82,252,330 listings, or (80.7%) have a blank job. Only 19,640,478 (19.3%) have one filled in.

This means that only 4 out of 5 listings have no job title. This means that a feature built on job titles would definitely miss most of the data.

4. Limits

- The limits of this are that job titles are often shown as /N which means they are blank and they are not normalized.
- It is also gonna be convenient to parse since the columns are split. They are split by tabs so the cut command can work more reliably.
- The file also includes other stuff such as general TV, and also shorts, games, etc. The next sprint were planning on filtering
- The numbers are counting listings and they are not counting unique people.

5. Timing

Step	Time
Count rows	3 s
Table 1 (role type)	1 min 16 s
Table 2 (job title)	1 min 13 s
Top 10	1 min 43 s
Filter	27 s
Skinny table	1 min

The full run took about 6 minutes on 101.9 million rows.

6. Links

- Repo: <https://github.com/sjsu-cs131-f26/CineMatch-Netflix-Movies>
- PRs: <https://github.com/sjsu-cs131-f26/CineMatch-Netflix-Movies/pull/1>,
<https://github.com/sjsu-cs131-f26/CineMatch-Netflix-Movies/pull/2>
- Board: <https://cs131-project.atlassian.net/jira/core/projects/CSPROJ/board>

7. Who did what

Member	Sprint 1 → 2	Contribution
Hadi Khan	Engineer → PM	Board, question, tables, filter, skinny table
Krishna Kuchibhatla	Storyteller → Engineer	Data card
Prajwal Dacharla	Storyteller → Engineer	First run script
Aneesh Matcha	PM → Storyteller	Report
Pouya Anvari	Engineer → Storyteller	Dataset, profile, sample, script fixes, report

8. AI use

Claude helped us explain commands and errors when we were running out errors. Claude and other AI chatbots also helped us fix the run script. However Claude did not help with development of this report, only for scaffolding and structure.. All commands were run and verified by hand.

9. Stakeholder update

Over 80% of 100+ million IMDb listings do not include a job title; however every job listing is showing a role type. CineMatch (we) should match people by the role type and not the job title and this is the plan. For the next sprint we're only gonna filter by movies.